

Summary

Detail-oriented Mechanical Engineering undergraduate at NIT Hamirpur with hands-on experience in lean manufacturing, line balancing, and process optimization. Proficient in CAD tools (SolidWorks, AutoCAD), ANSYS (FEA), and working knowledge of CFD, stress analysis, and prototyping. Experienced in applying engineering methods such as Design for Manufacturability (DFM), Design for Assembly (DFA), Value Engineering, Concurrent Engineering, and Statistical Process Control (SPC) to improve efficiency, reduce costs, and enhance product quality. Adept at implementing root cause analysis, quality assurance, and continuous improvement techniques. Strong communicator and collaborative team player seeking to contribute technical insight to manufacturing or design-focused roles.

Skills

Solidworks(3D modelling)

CAD design)

ANSYS (FEA for thermal & stress analysis)

CNC Machining – programming

Toolpath Verification

MS Excel – BOM management

data analysis

Production Planning

AutoCAD(3D/2D)

Soft Skills

Analytical Thinking · Problem-Solving · Team Collaboration · Project Management · Communication Skills

Education

NATIONAL INSTITUTE OF TECHNOLOGY HAMIRPUR

Bachelor of Technology - Mechanical Engineering, CGPA: 8.34

St. Mary's School Bijnor

Class 10; ICSE, CGPA: 9.6

2019

St. Mary's School Bijnor

Class 12; ISC, CGPA: 9.6

2021

Experience

Moon Beverages Ltd. (Coca-Cola)

Intern- Production Department

Dec 2024 - Jan 2025

Dadri, Uttar Pradesh

- Assisted in daily operations related to manufacturing and production planning
- Observed and learned key aspects of industrial workflows and plant coordination
- Built communication and adaptability skills in a high-paced environment
- Supported preventive maintenance planning and quality checks—helped reduce machine downtime by ~5%

Nature Green Tools And Machines Pvt. Ltd.

Intern – CNC Machine Operations & Production

May 2025 - Jun 2025

Greater Noida, Uttar Pradesh

- Worked with CNC machines for programming, operation, and maintenance
- Supported process improvement efforts to increase production efficiency

- Supported preventive maintenance planning and quality checks—helped reduce machine downtime by ~5%.
- Introduced SMED-based tool change optimizations, improving machine uptime by ~10%.
- Identified defect causes and reduced scrap by ~8% through root cause analysis

Personal Project

Project: Line Balancing of HPNC Assembly Line

- Optimized crane assembly line by identifying bottlenecks and reducing idle time
- Redesigned task distribution and workstation layout to boost throughput by ~12%.
- Improved workflow, productivity, and layout by applying lean and industrial engineering concepts
- Implemented lean tools (5S, Kaizen), created technical documentation for task layout and assembly flow
- Studied cycle and takt time, incorporated feedback loops for continuous process improvement

Leadership & Extra Curriculars

- **Executive Member**, SRIJAN Club, NIT Hamirpur
Helped organize major cultural and technical events; built leadership and coordination skills
- **Cricket Team Member**, NIT Hamirpur
Represented college in the **All India Inter-NIT Tournament**
Winner – Inter-Branch | Runner-up – Inter-Year tournaments

Relevant Coursework

Manufacturing Processes · Thermodynamics · Heat and Mass Transfer · Strength of Materials · Industrial Engineering · Machine Design · Fluid Mechanics · Engineering Mechanics · Mechanics of Materials · Production Planning and Control